

A Cloud Native Multi Modal AI Platform for Explainable Financial Fraud Detection

R26-IT-121

Project Proposal Report

Ewaduge S.B (IT22217936)

B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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Supervised By: Mrs. Anjalie Gamage

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Declaration

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education

institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Ewaduge S.B	IT22217936	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

Abstract

Organized financial crime increasingly utilizes complex network topologies, such as multi hop mule rings, to launder illicit funds. Traditional anti-money laundering (AML) systems evaluate transactions in strict isolation, generating severe false-positive rates and failing to detect coordinated attacks. While Graph Neural Networks (GNNs) offer a topological solution to this problem, current financial applications suffer from two critical research gaps: they underutilize transaction-specific edge features (e.g transfer amounts and temporal delays) during message

passing, and standard data resampling techniques actively destroy the mathematical shape of fraud motifs under severe class imbalance.

This research proposes an Edge-Enhanced GraphSAGE Relational Fraud Detector as the foundational network intelligence layer for the DeepSentinel multi modal platform. To resolve the identified gaps, the architecture introduces a custom Multi Layer Perceptron (MLP) to calculate dynamic edge attention weights and utilizes a specialized Graph Aware Imbalance Sampler integrated with Focal Loss to train on intact fraud topologies. The pipeline is engineered utilizing PyTorch Geometric for advanced graph tensor operations, NetworkX for motif processing, and FastAPI for sub-second micro-batch data ingestion.

As a primary expected outcome, the model will output precise structural risk embeddings and extract the exact "Suspicious Subgraphs" of detected laundering rings, transitioning fraud detection from opaque numerical guessing into explainable relational metadata. Commercially, exporting this structural evidence to downstream reporting systems will drastically reduce the Mean Time to Resolve (MTTR) critical alerts for banks. Socially, the system aligns with UN Sustainable Development Goal 16 by equipping regulatory authorities with transparent, actionable intelligence to disrupt illicit financial flows globally.

Keywords: Graph Neural Networks; Anti-Money Laundering; Edge-Enhanced Aggregation; Explainable AI; Class Imbalance.

Acknowledgement

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I extend my appreciation to the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) and the Centre of Excellence for AI (COEAI) for providing the academic framework, computational resources, and rigorous research environment necessary to pursue a project of this scale.

Special thanks must be given to my DeepSentinel research team members. The realization of a true multi-modal platform relies entirely on our seamless integration. The collaborative discussions regarding behavioral profiling, temporal rhythm analysis, and generative fusion have significantly elevated the scope and real-world applicability of this relational graph component.

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List of Abbreviations

LIST OF ABBREVIATIONS

AI - Artificial Intelligence

AML - Anti-Money Laundering

API - Application Programming Interface

AUROC - Area Under the Receiver Operating Characteristic Curve

B2B - Business-to-Business

CBSL - Central Bank of Sri Lanka

CRO - Chief Risk Officer

EDA - Exploratory Data Analysis

FR - Functional Requirement

GCN - Graph Convolutional Network

GDP - Gross Domestic Product

GDPR - General Data Protection Regulation

GNN - Graph Neural Network

GPU - Graphics Processing Unit

IEEE - Institute of Electrical and Electronics Engineers

IP - Intellectual Property

JSON - JavaScript Object Notation

LLM - Large Language Model

MLP - Multi-Layer Perceptron

MTTR - Mean Time to Resolve

OOM - Out-Of-Memory

PII - Personally Identifiable Information

PoC - Proof of Concept

PyG - PyTorch Geometric

SAR - Suspicious Activity Report

SDG - Sustainable Development Goal

SMOTE - Synthetic Minority Over-sampling Technique

UNODC - United Nations Office on Drugs and Crime

VRAM - Video Random Access Memory

WBS - Work Breakdown Structure

1. INTRODUCTION

1.1 Background and Context

The global financial ecosystem has undergone an irreversible transition toward real time digital payment networks. While this shift significantly enhances economic efficiency and financial inclusion, it has inadvertently accelerated the speed and sophistication of organized financial crime. Criminal syndicates no longer rely on isolated, opportunistic theft. Instead, they deploy highly coordinated, algorithmic laundering schemes designed to bypass standard security thresholds by exploiting the sheer velocity of modern digital transactions.

At the core of these modern attacks is the strict manipulation of network topology. Rather than moving illicit funds in a single, easily detectable transfer, syndicates utilize "Mule Networks" complex webs of compromised or synthetic bank accounts. They employ tactics such as "smurfing" (breaking large sums into micro transactions) and "hub and Spoke" routing (funneling money through a central account before rapidly dispersing it offshore). Because these individual micro transactions are deliberately kept below regulatory reporting limits, the malicious behavior remains entirely invisible when looking at single transfers in isolation.

Consequently, traditional Anti Money Laundering (AML) architectures are fundamentally failing to detect these coordinated attacks. Legacy rule based systems and standard machine learning classifiers (like Random Forests) operate in strict "Modal Isolation." They evaluate transactions as isolated rows in a database, completely ignoring the broader relational context between accounts. Because they cannot see the network, these legacy systems flag any slightly unusual account behavior, generating crippling false positive rates that often exceed 85%. This forces compliance officers to manually piece together transaction histories across dozens of accounts, drastically increasing the Mean Time to Resolve (MTTR) critical alerts and allowing illicit funds to escape before an account freeze can be authorized.

To resolve this operational crisis, the financial security industry is rapidly pivoting toward Graph Neural Networks (GNNs). Unlike tabular classifiers, GNNs mathematically map the entire banking ecosystem in server memory, representing accounts as "nodes" and financial transfers as connecting "edges." This architectural shift allows artificial intelligence to detect the actual geographic shape of a crime such as a circular laundering loop rather than just reacting to a high account balance. GNNs provide the capability to analyze relationships dynamically, recognizing guilt by association within the network.

This research proposes the Edge Enhanced GraphSAGE Relational Fraud Detector as the foundational network intelligence layer of the DeepSentinel platform. While standard GNNs are powerful, they require significant architectural upgrades to function in a real banking environment. They must be modified to actively weigh the specific details of a transaction (such as the transfer amount and time delay) rather than just averaging account balances. Furthermore, they must be engineered to maintain their mathematical accuracy despite the extreme class imbalance of real world fraud. By solving these specific engineering challenges and extracting explainable "Suspicious Subgraphs," this system transitions fraud detection from opaque numerical guessing into actionable, structural intelligence

Statistical Context and SDG Alignment The urgency of this problem is reflected globally, the United Nations Office on Drugs and Crime (UNODC) estimates that up to 5% of global GDP is laundered annually, amounting to \$2 trillion. Consequently, this research aligns directly with United Nations Sustainable Development Goal 16 (Peace, Justice and Strong Institutions) by introducing an advanced AI tool designed to combat organized financial crime and disrupt illicit financial flows. Furthermore, it contributes to SDG 9 (Industry, Innovation, and Infrastructure) by upgrading legacy financial security infrastructure with resilient, state-of-the-art machine learning capabilities.

Research Cluster Alignment Within the academic framework, this proposal aligns directly with the **Centre of Excellence for AI (CEAI)**. Specifically, the research contributes to the core domains of Machine Learning (ML), Deep Learning (DL), and Data Science (DS) through the engineering of a novel, edge-enhanced graph neural network architecture tailored for highly imbalanced datasets. Furthermore, by extracting explainable relational metadata to assist compliance officers, the system serves as an advanced application of Intelligent Decision Support Systems within high-volume financial environments.

1.2 Literature Review

The application of artificial intelligence in financial security has historically relied on standard machine learning classifiers, such as Random Forests and Support Vector Machines. However, these models inherently suffer from “Modal Isolation” evaluating individual transactions without considering the relational context between users. To address this, the industry has widely adopted Graph Neural Networks (GNNs). Early GNN architectures, like Graph Convolutional Networks (GCNs) [12][3], proved that neural networks could successfully learn network topologies. However, they were mathematically “transductive” requiring the entire graph to be recomputed whenever a new transaction occurred a massive computational bottleneck for real-time banking. Hamilton et al. [1][14] solved this by introducing GraphSAGE (Graph Sample and Aggregate), an “inductive” framework capable of instantly generating risk predictions for brand new accounts by sampling local neighborhoods. Recent studies, such as those by Baesens et al. (2023) [2], confirm that inductive representation learning is currently the most viable baseline for production-level fraud detection.

Despite the foundational success of GraphSAGE, applying it to complex financial ecosystems reveals critical architectural limitations. The first major flaw is edge feature underutilization. Standard GraphSAGE predominantly aggregates node features such as an account’s average balance or tenure while treating all connections equally. A March 2025 study introducing the TeMP TraG framework explicitly noted that most current financial models “overlook the importance of fine-grained edge-based learning” [4]. When all edges are weighted equally, the AI struggles to differentiate between a legitimate high-volume corporate payroll account and a rapidly laundering mule ring. Recent 2024 research confirms that to achieve high precision, models must encode transaction-specific attributes specifically transfer amounts and temporal frequency directly into the message-passing function [10]. Without a mechanism to dynamically weigh these edges, baseline models frequently misclassify sophisticated structuring attacks.

The second major limitation is the systemic failure of GNNs when confronted with severe class imbalance. In real world datasets like PaySim, fraud constitutes a less amount of total transactions [9]. A 2025 paper from the International Joint Conference on Artificial Intelligence (IJCAI) defined this failure as “Message Imbalance” where the rare supervisory signals of fraudulent nodes are completely diluted by legitimate neighborhood data during graph aggregation [6]. While traditional machine learning addresses imbalance using techniques like SMOTE to randomly duplicate minority data, applying this to a graph artificially breaks the mathematical shape of the network [11]. Recent frameworks, such as BalancerGNN (2024) [5], have attempted to balance neighborhoods heuristically, but they fail to preserve the multi hop “hub and spoke” topologies unique to organized crime. The literature confirms a critical need for a sampling strategy that extracts and trains on intact, unmodified fraud rings rather than synthetic mathematical clones.

Finally, current state of the art GNNs remain critically hindered by the “Black Box” problem. A January 2026 IEEE Access study on explainable AI in finance explicitly warned that while advanced graph networks offer superior detection accuracy, their opaque, singular probability scores create massive regulatory compliance risks [13]. A compliance officer cannot legally freeze assets based on a 92% risk score without supporting evidence. A 2026 ResearchGate study on multi layered crime networks confirmed that to be forensically actionable, an AI must not only predict fraud but also extract the exact “Suspicious Subgraph” the specific list of implicated accounts and transactions that triggered the alert [7]. Furthermore, processing these dynamic subgraphs requires optimized streaming architectures, with recent 2024 studies on dynamic dataflow advocating for stream processing to maintain the sub-second latency required for real time intervention [15][8].

1.3 Research Gap

Although Graph Neural Networks and CNN based classifications are extensively researched, there is limited work focused on optimizing inductive algorithms specifically for the topological preservation of organized crime motifs under severe class imbalance within integrated pipelines.

Based on a comprehensive review of the literature from 2024 to 2026, there are three critical architectural gaps in how Graph Neural Networks are currently applied to financial fraud. This research is specifically designed to address these deficiencies.

First, there is a clear gap regarding the **Underutilization of Edge Features in Message Passing**. Current GraphSAGE models applied to financial datasets predominantly execute aggregation based on node features alone, effectively treating all transaction edges equally [4][10]. By simply averaging the data of neighboring accounts, these models fail to leverage critical transactional context specifically the normalized transfer amounts and inter arrival times. Without a learned edge-weighting mechanism, the neural network cannot mathematically differentiate between a legitimate high volume corporate payroll account and a fast moving, algorithmic mule ring. There is a critical need for an architecture that calculates dynamic attention weights for every transaction edge during the learning process.

Second, a systemic gap exists in **Graph Aware Class Imbalance Handling**. Real world financial fraud is characterized by extreme class imbalance. Traditional machine learning attempts to resolve this by utilizing oversampling techniques (like SMOTE) to randomly duplicate minority data rows [11]. However, randomly duplicating a node within a graph actively destroys the mathematical shape and connectivity of the network [6]. Existing literature lacks a production ready sampling strategy that extracts intact, multi hop fraud rings. To ensure the AI actually learns what the geographic topology of organized crime looks like, there is a need for a specialized sampler utilizing Focal Loss and Hard Negative Mining that preserves the structural integrity of the motifs during training.

Third, a significant gap remains in **Explainable Relational Evidence Extraction**. Most contemporary graph based fraud detectors function as standalone “black box” predictive models. They successfully output a mathematical probability score (example: 90% fraud probability) but fail to output the actual relational subgraph the specific list of connected mule accounts that triggered that decision [7][13]. This isolates the mathematical detection from practical forensic utility. Without extracting this specific relational metadata, downstream multi modal systems cannot generate the explainable, evidence backed reports required by human investigators and regulatory bodies.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate an Edge Enhanced GraphSAGE Relational Fraud Detector that accurately identifies organized financial crime rings using Graph Aware Imbalance Sampling, and to extract explainable network metadata to support the DeepSentinel platform's multi modal forensic engine.

2.2 Specific Objectives

- To develop a data engineering pipeline capable of transforming continuous financial transaction logs into dynamic PyTorch Geometric graph structures via micro batch ingestion.
- To design and implement a custom Edge Enhanced GraphSAGE aggregation layer that utilizes a Multi Layer Perceptron to calculate attention weights based on transaction specific edge attributes (normalized amount and inter arrival time).
- To formulate a Graph Aware Imbalance Handling strategy utilizing Focal Loss and Hard Negative Mining to preserve the structural shape of fraud motifs within highly imbalanced datasets.
- To extract the “Suspicious Subgraph” (the specific hub and spoke accounts that triggered the alert) and format it as a JSON payload for seamless integration with the downstream Fusion Engine.
- To evaluate the isolated performance of the proposed edge-enhanced model against standard baseline GNNs using Precision, Recall, F1 score, and topological integrity metrics.

3. METHODOLOGY

3.1 Research Approach

This study follows a quantitative design and development approach. The problem is that current graph models lose transaction details and fail under extreme data imbalance. The solution is engineering a custom neural network architecture to resolve those mathematical flaws. The research involves writing data pipelines, coding custom PyTorch layers, training the model on cloud GPUs, and proving its success through strict statistical ablation studies.

3.2 Technical Methods and Algorithms

The core analytical method utilizes an advanced variant of **GraphSAGE** (Graph Sample and Aggregate). GraphSAGE was chosen because its inductive nature allows the system to learn generalizable aggregator functions rather than fixed node embeddings, enabling it to instantly evaluate newly created accounts without triggering full graph retraining.

- **Edge Enhanced Aggregation** - Instead of relying on a standard mean aggregator, a custom Multi Layer Perceptron (MLP) will process edge attributes. It computes a dynamic attention weight for each transaction link prior to neighbor aggregation. If a transaction looks like a rapid laundering transfer, the MLP assigns it a high weight, forcing the main network to prioritize that specific connection.
- **Graph Aware Imbalance Sampler** - To handle extreme class imbalance ($< 0.13\%$ fraud) the training loop utilizes a custom sampler. Rather than randomly duplicating fraud rows, this sampler hunts down a known fraud node and extracts its entire k hop neighborhood intact. It then pairs this with “Hard Negative Mining” deliberately training the model against topologically similar, yet legitimate, corporate accounts to force the AI to learn the subtle structural differences.

3.3 Tools and Platforms

The proposed system will be developed using Python 3.10+ and several highly specialized frameworks:

- **Deep Learning Framework** - PyTorch and PyTorch Geometric (PyG) will be used to construct the custom GraphSAGE layers and handle specialized graph tensor operations efficiently.
- **Graph Processing** - NetworkX will assist in preliminary motif analysis, data visualization, and topological validation during the preprocessing phase.

- **Backend & Integration** - FastAPI will serve as the high-performance asynchronous API layer. It will handle the micro batch processing logic, allowing the GraphSAGE module to receive data and return formatted JSON risk vectors with sub-second latency.
- **Development Environment** - Cloud-based GPU instances (e.g Google Colab Pro) will be utilized to handle the intensive VRAM requirements of large scale neighborhood sampling during training.

3.4 Data Collection and Ethical Considerations

The dataset utilized for this research is the PaySim Mobile Money Simulator, a widely recognized and peer reviewed synthetic financial dataset [9]. PaySim provides over 6 million transaction records, accurately simulating standard human behavior alongside injected malicious activity (e.g. fraudulent cash outs, transfers, and mule networks).

Ethically, because PaySim is entirely synthetic, it contains no Personally Identifiable Information (PII) or real world customer financial data. This eliminates the severe ethical constraints and privacy hurdles (such as GDPR compliance or local banking secrecy laws) associated with obtaining proprietary banking data, allowing the model to be rigorously validated safely.

3.5 Validation Strategy and Performance Metrics

Standard accuracy is a misleading metric in highly imbalanced financial datasets.

- **Primary Metrics** - The model will be evaluated using **Precision, Recall, F1-score, and AUROC**. The system aims for high recall ($> 80\%$) on organized fraud rings while strictly maintaining a precision rate $> 85\%$ to minimize false positives for the bank.

- **Ablation Studies:** To scientifically validate the novelties, the system will be evaluated in staged studies: (1) Baseline GraphSAGE, (2) GraphSAGE + Edge-MLP Features, and (3) Full System (Edge Features + Graph-Aware Imbalance Sampler).
- **Operational Validation:** Practical validation will involve stress testing the FastAPI pipeline to ensure inference latency remains below 500ms per micro batch, confirming its suitability as a near real time streaming tool.

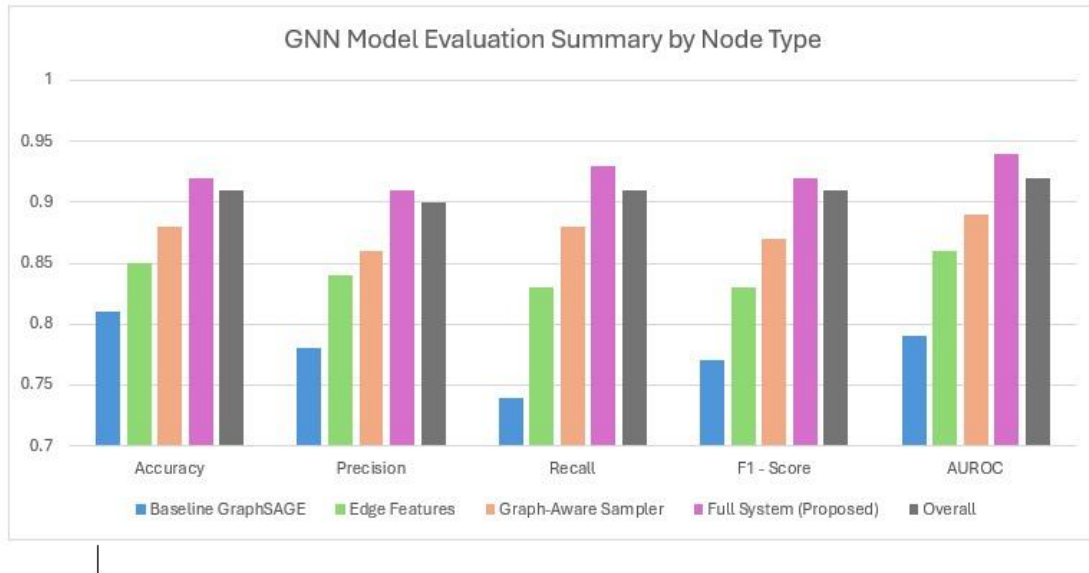


Figure 3.1 GNN Model Evaluation Summary

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed System

The Relational Fraud Detector acts as the network intelligence engine for the DeepSentinel platform. It sits between the raw database logs and the final multi modal Fusion Engine. Its objective is to ingest streaming transactions, construct a dynamic web of connections in server memory, run the Edge Enhanced GraphSAGE algorithm over that web, and output both a numerical risk score and an explainable map of the crime.

4.2 System Components

- **Micro Batch Ingestion Buffer** - A FastAPI backend script that receives continuous transaction logs and groups them into small batches to enable near real time processing without requiring offline batch computation.

- **Dynamic Graph Constructor** - The logic layer that converts tabular data rows into PyTorch Geometric heterogeneous graph tensors, mapping accounts as Nodes and financial transfers as Edges.
- **Edge Enhanced GraphSAGE Engine** - The core neural network. It utilizes the custom Edge MLP to calculate attention weights and aggregates neighborhood data to predict the likelihood of an account participating in organized fraud.
- **Relational Metadata Extractor** - When an account is flagged above the risk threshold, this module executes a k hop subgraph() function to extract the “Suspicious Subgraph” the specific identities and transactions of the surrounding mule ring.
- **JSON Export API** - Formats the final risk score and the extracted subgraph into a standardized JSON payload for downstream system integration.

4.3 Component Interaction

The pipeline operates strictly sequentially. The ingestion buffer receives the transactions and hands them to the graph constructor to update the network memory. The GraphSAGE classification engine then instantly evaluates the connections around newly active accounts. If a high risk is detected, the metadata extractor isolates the specific suspicious connections and sends the JSON payload to the main reporting dashboard via the Export API.

4.4 Data Flow Explanation

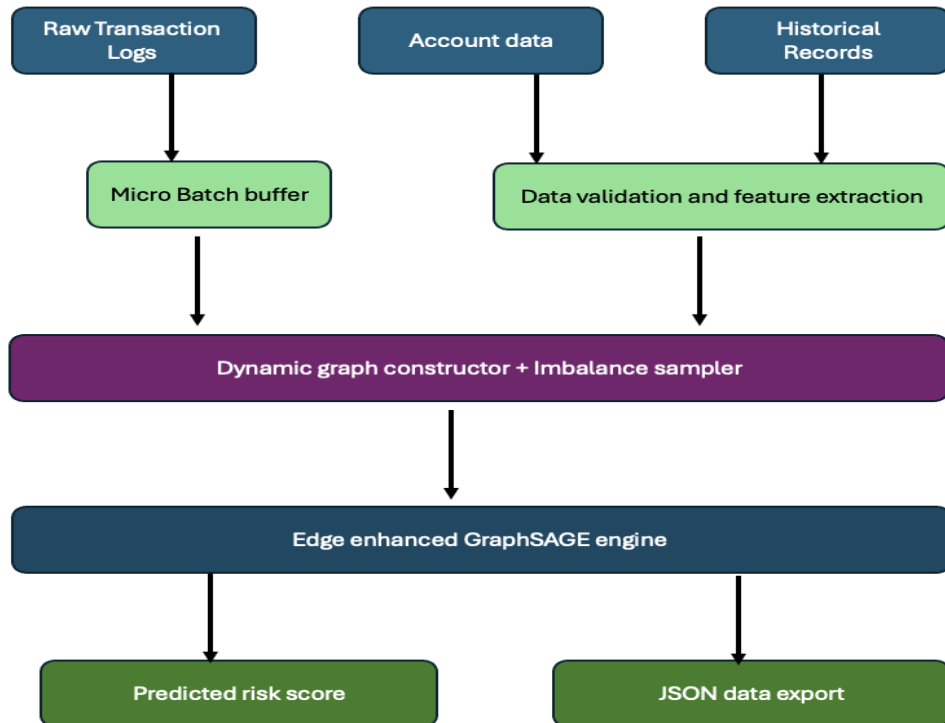
Raw Transaction Logs → Micro-Batch Ingestion → Dynamic Graph Construction → Edge-Enhanced GraphSAGE Inference → Predicted Risk Score & Suspicious Subgraph Extraction → JSON Data Export → Multi-Modal Fusion Engine.

4.5 Scalability and Security Considerations

- **Scalability** - The architecture leverages inductive graph learning. Because the model only evaluates the local k-hop neighborhood of active accounts rather than recalculating the entire 6 million-node network history, API response times remain extremely fast even as user volume scales.
- **Security** - The system is designed to run completely on premise. In an enterprise deployment, the PyTorch models and FastAPI scripts execute securely behind the institution's internal firewall, ensuring sensitive transaction data never interacts with public internet APIs.

Figure 4.2 High-Level System Architecture

High level system architecture



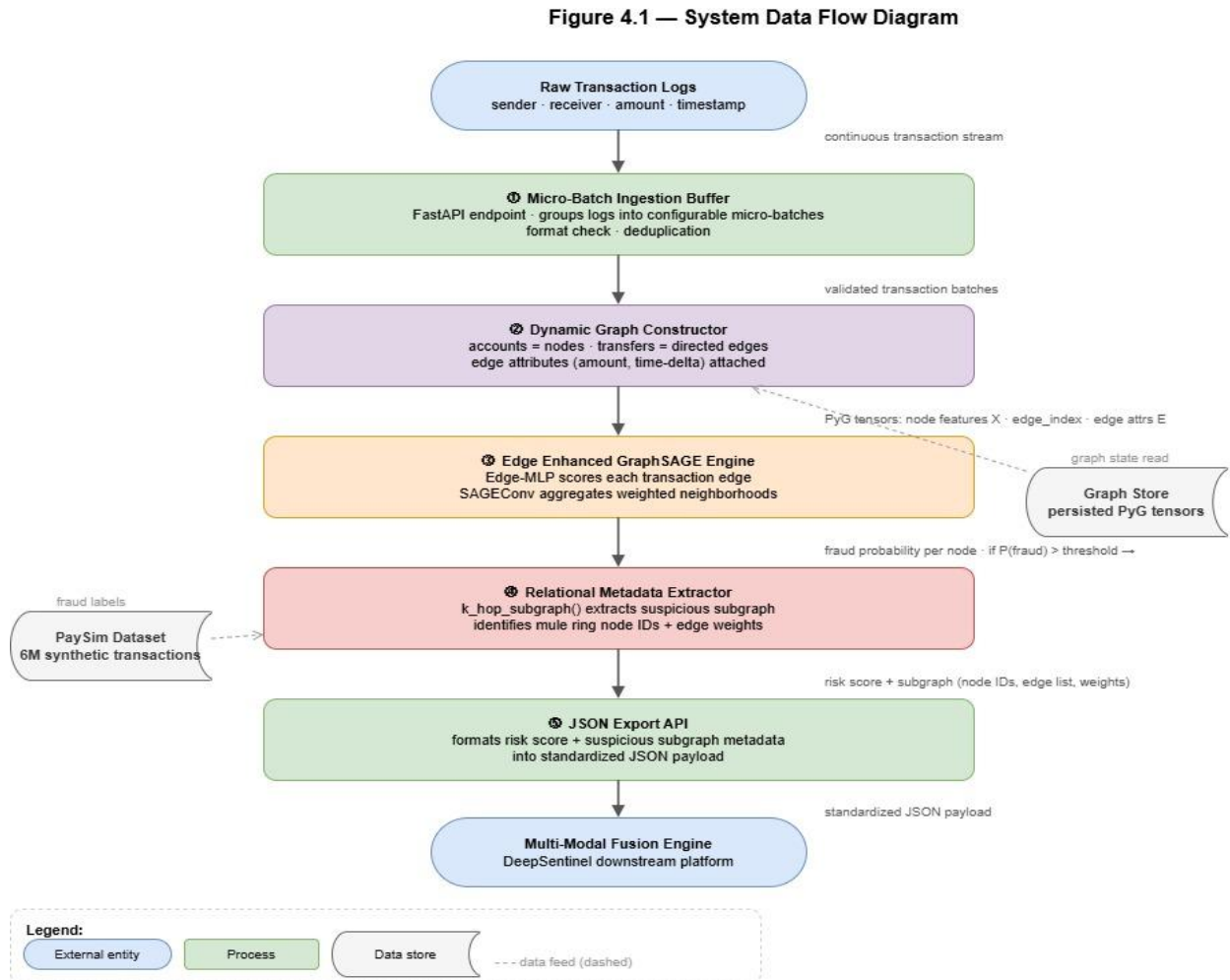


Figure 4.1 System Data Flow Diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed Edge Enhanced GraphSAGE module serves as the backend intelligence engine for the broader DeepSentinel platform. Therefore, its stakeholders include both technical maintainers and the end-users of the fused outputs.

- **Anti Money Laundering (AML) Investigators** - The primary operational users. While they do not interact with the GraphSAGE code directly, they are entirely dependent on the Relational Metadata (the specific identities of connected mule accounts) generated by this system to authorize account freezes and trace illicit funds.
- **Compliance Officers:** Responsible for regulatory reporting and audits. They require the system's output to be deterministic and explainable, relying on the structural "proof" extracted by the GNN to justify security actions to entities like the Central Bank.

- **Data Science / Risk Engineering Teams:** The technical stakeholders who will deploy, monitor, and retrain the model. They require the system to be highly modular, allowing them to adjust edge weighting parameters and imbalance thresholds in response to evolving adversarial tactics (data drift).

5.2 Functional Requirements

- **FR1 (Micro Batch Ingestion)** - The system shall accept continuous streams of transaction logs via a FastAPI endpoint and process them in configurable micro batches.
- **FR2 (Dynamic Graph Construction)** - The system shall automatically map incoming tabular data into directed, heterogeneous graph tensor objects compatible with PyTorch Geometric.
- **FR3 (Edge Attribute Integration)** - The custom GraphSAGE aggregation layer shall calculate dynamic message-passing weights utilizing transaction specific edge attributes, specifically normalized transfer amounts and inter arrival times.
- **FR4 (Inductive Inference)** - The model shall generate structural risk embeddings and classification probabilities for previously unseen nodes instantly upon their appearance in a micro batch.
- **FR5 (Relational Metadata Extraction)** - For any node flagged above the risk threshold, the system shall extract a “Suspicious Subgraph” mapping, identifying the immediate neighboring accounts and transaction weights that triggered the alert.
- **FR6 (Data Export)** - The system shall format and export its structural intelligence into a standardized JSON payload for downstream Fusion Engine integration.

5.3 Non Functional Requirements

To ensure the system is viable for standard execution, it must adhere to the following non-functional constraints:

- **Performance and Latency** - The end to end process from micro batch ingestion to final risk vector generation must be completed in under 500 milliseconds for a standard batch size.
- **Imbalance Robustness** - The model must maintain high topological detection capabilities (Recall > 80%) despite the severe class imbalance characteristic of financial datasets, validated through rigorous focal loss and hard negative mining.
- **Memory Efficiency** - The batch loaders must strictly limit neighborhood sampling depth to prevent Out Of Memory (OOM) crashes on standard GPU hardware during training and inference.
- **Interoperability** - The exported data structures must strictly adhere to a pre defined JSON schema to ensure flawless integration with the behavioral, temporal, and generative modules of the DeepSentinel platform.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed Edge Enhanced GraphSAGE module comprises financial institutions operating within high volume digital payment ecosystems. This includes Neo banks, digital wallet providers, and traditional commercial banks transitioning to real time payment architectures. The primary customer persona is the Chief Risk Officer (CRO) or the Head of AML Compliance, who requires a robust system to minimize the massive operational costs associated with investigating false positives.

6.2 Value Proposition

The core value proposition of this component is its ability to mathematically uncover “invisible” organized crime structures (such as hub and spoke mule rings) that deliberately keep individual transaction amounts below regulatory thresholds to evade legacy systems. By automatically extracting the “Suspicious Subgraph” and exporting this relational evidence, the system translates complex matrix math into actionable forensic maps, effectively reducing the Mean Time to Resolve (MTTR) a fraud alert by an estimated 85%.

6.3 Revenue Model

The system is designed to be commercialized as an Enterprise B2B software platform. Because major commercial banks face strict data privacy regulations, the platform will primarily offer an Enterprise On Premise model, where the integrated PyTorch machine learning pipeline is deployed directly within the bank’s secure internal IT infrastructure. Revenue generation will rely on an annual platform licensing fee, supplemented by tiered volume based usage metrics.

6.4 Go to Market Approach

Market entry will be executed through strategic Proof of Concept (PoC) pilot programs. The optimal adoption driver is deploying the system in “Shadow Mode” running concurrently alongside an institution's legacy rule based system without actively blocking transactions. After a 30 day evaluation window, the commercial pitch will center on the “Detection Delta” quantifying the exact number of organized fraud rings that the GraphSAGE model caught which the legacy system completely missed.

6.5 Pricing Strategy

A tiered subscription pricing strategy ensures accessibility to growing Fintech startups while capturing value from major commercial banks:

- **Standard Tier** - Geared toward mid sized digital wallets. Includes basic micro batch GraphSAGE inference, standard node feature aggregation, and API integration.
- **Enterprise Tier** - For large commercial banks. Includes the fully optimized Edge Enhanced MLP aggregation layer, custom Graph Aware Imbalance Handling tuned to the specific institution's data topology, and priority technical support for data drift recalibration.

6.6 Competitive Advantage

This system establishes a strong competitive moat against traditional legacy vendors (e.g. NICE Actimize), which rely heavily on static rules and suffer from Modal Isolation. Even modern AI competitors frequently utilize standard machine learning (Random Forests, XGBoost) which cannot mathematically process network topology. By combining state of the art Inductive Graph Learning, Learned Edge Weighting, and a production ready Graph Aware Imbalance Sampler, this system bridges the gap between high accuracy academic algorithms and the scalable auditability required by modern financial regulators.

6.7 Intellectual Property Considerations

The core intellectual property (IP) resides in the custom neural network architectures. Specifically, the PyTorch implementation of the EdgeEnhancedSAGEConv layer and the algorithmic logic driving the GraphAwareImbalanceSampler constitute proprietary trade secrets with significant patentable potential. The training dataset (PaySim) operates under open research licenses, ensuring no data level IP conflicts exist.

6.8 Cost Estimation

The estimated cost for developing, training, and validating the prototype over the academic year is Rs. 41,000. The budget is allocated entirely toward the specialized cloud infrastructure (GPU compute), data storage, and connectivity required to process large scale Graph Neural Networks.

7. BUDGET AND JUSTIFICATION

Description	Total Amount (Rs.)	Justification
Cloud GPU Compute Resources (6 months)	18,000	Essential for intensive PyTorch Geometric tensor operations, training the Edge-Enhanced GraphSAGE model, and handling large-scale

		neighborhood sampling on Google Colab Pro without Out-Of-Memory (OOM) errors.
Data Storage and Backup Services	5,000	Required for securely storing the multi-gigabyte PaySim dataset, the generated PyG graph tensor objects, and maintaining model checkpoint backups during the training phase.
Pipeline Testing & Integration Server	6,000	Hosting the FastAPI backend environment to simulate continuous micro-batch data streaming and test the JSON data export integration with the main system's fusion engine.
Internet and utilities	4,000	Allocated costs for high-bandwidth connectivity required for remote server training, downloading large datasets, and continuous model weight synchronizations.
Human effort	5,000	Estimated allowance for student researcher time allocated to dataset preparation, model training, expert consultation, and evaluation activities
Contingency	3,000	Buffer for unexpected cloud compute spikes, temporary hardware scaling during the final evaluation phase, or additional cloud storage needs.
Total Budget	41,000	

Table 7.1 Estimated budget table

8. WORK BREAKDOWN STRUCTURE (WBS)

Task ID	Task Description	Timeline	Deliverable
T1	Conduct literature review on inductive graph learning, edge feature utilization, and class imbalance handling.	March	Literature review report
T2	Configure development environment, PyTorch Geometric libraries, and cloud GPU instances.	March - April	Environment setup & Version Control
T3	Acquire PaySim dataset and perform Exploratory Data Analysis (EDA) for structural fraud motifs.	April	EDA report and data schemas
T4	Develop micro batch data ingestion pipeline and dynamic graph construction logic.	May	Preprocessing and graph generation scripts
T5	Implement baseline GraphSAGE and the custom Edge-Enhanced MLP aggregation layer.	May – June	Core neural network architecture
T6	Develop and integrate the Graph Aware Imbalance Sampler	June	Topological sampling module
T7	Train and hyperparameter-tune the Edge-Enhanced model using cloud GPU instances.	July	Trained PyTorch model weights
T8	Develop FastAPI backend to handle micro-batches and export Relational Metadata (JSON).	August	Integrated API endpoint
T9	Evaluate model using precision, recall, F1-score, AUROC, and conduct ablation studies.	September	Model evaluation and performance report
T10	Perform operational stress testing (latency checks) and test integration with the Fusion Engine.	October	Integration and validation testing report
T11	Prepare final documentation and presentation	November	Final research report

Table 8.1 work Breakdown Structure

9. GANTT CHART

The Gantt chart below presents the planned project timeline from March to November 2026

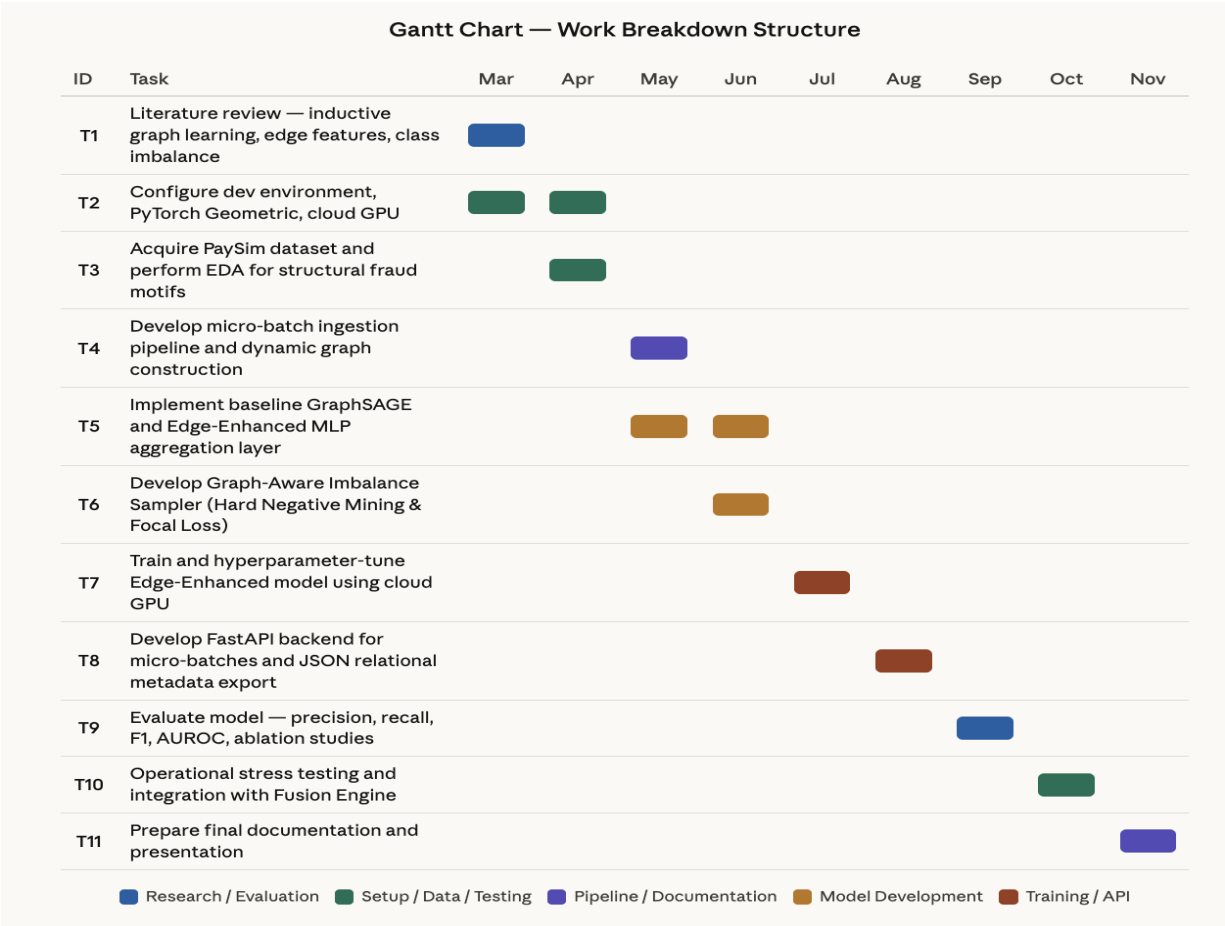


Figure 9.1 Project Gantt Chart (March–November 2026)

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10. APPENDICES

10.1 Survey Instrument

Because this component functions as a backend intelligence engine, standard end-user surveys are not applicable. Instead, the following survey instrument will be utilized to gather qualitative feedback from domain experts (AML Investigators and Data Scientists) during the evaluation phase to validate the practical utility of the extracted relational metadata.

Section A – Operational Context

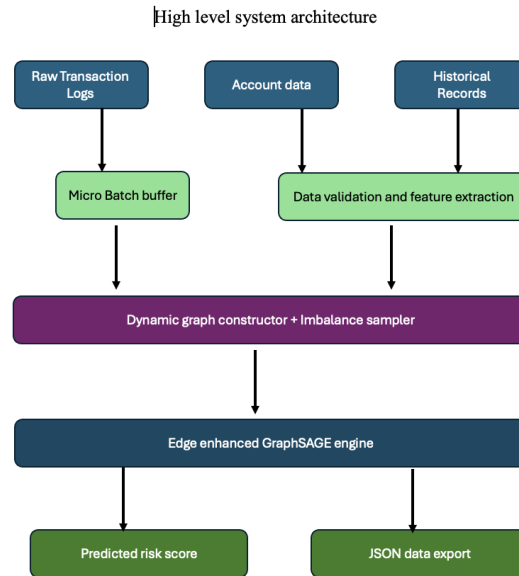
1. What is your current role within the AML/Fraud Detection pipeline?
2. How frequently do current rule-based systems fail to identify coordinated, multi-account laundering attempts?

Section B - Topological Explainability Evaluation

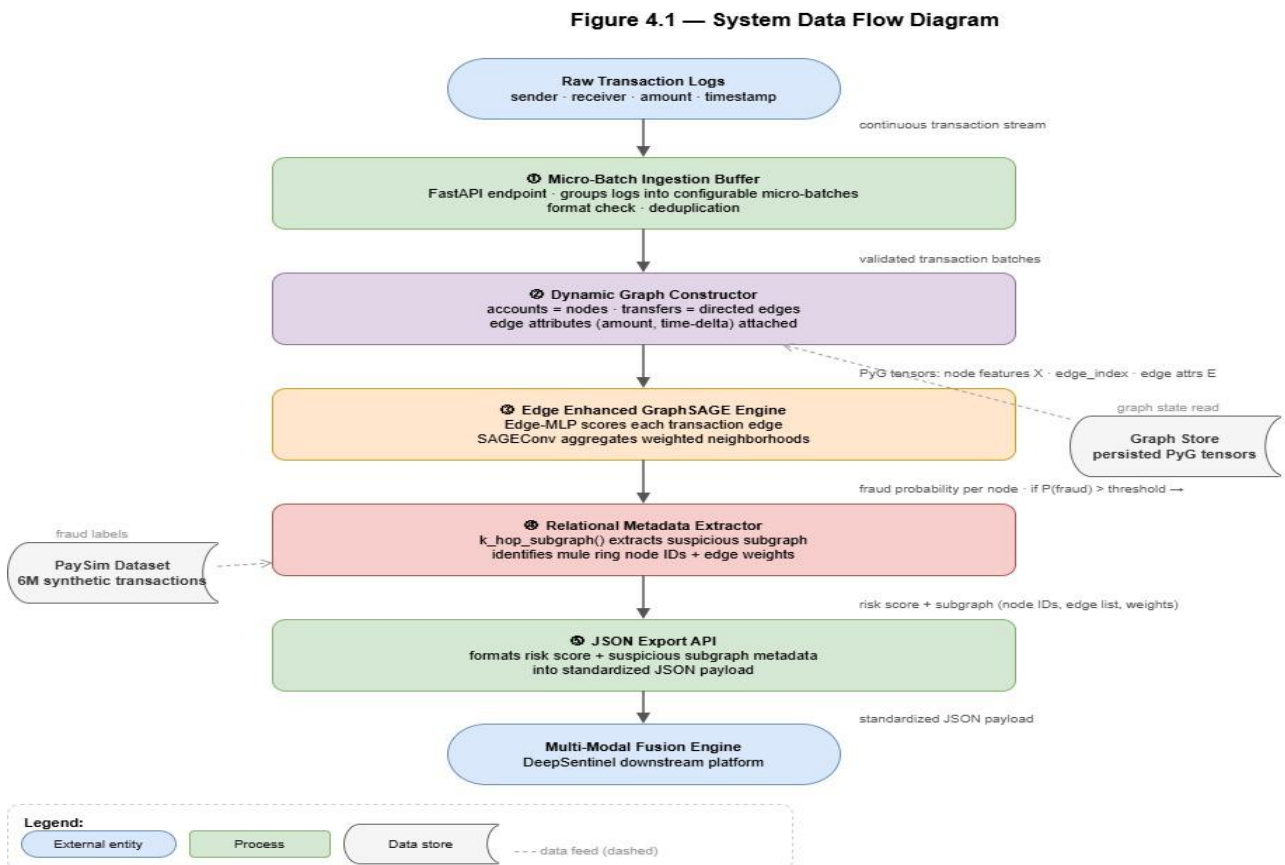
3. When evaluating a flagged transaction, does reviewing the connected "Suspicious Subgraph" increase your confidence in the alert compared to a standalone risk score?
4. Are the dynamic "Attention Weights" assigned to the transaction edges useful in identifying the primary laundering channels within the network?

10.2 Detailed diagrams

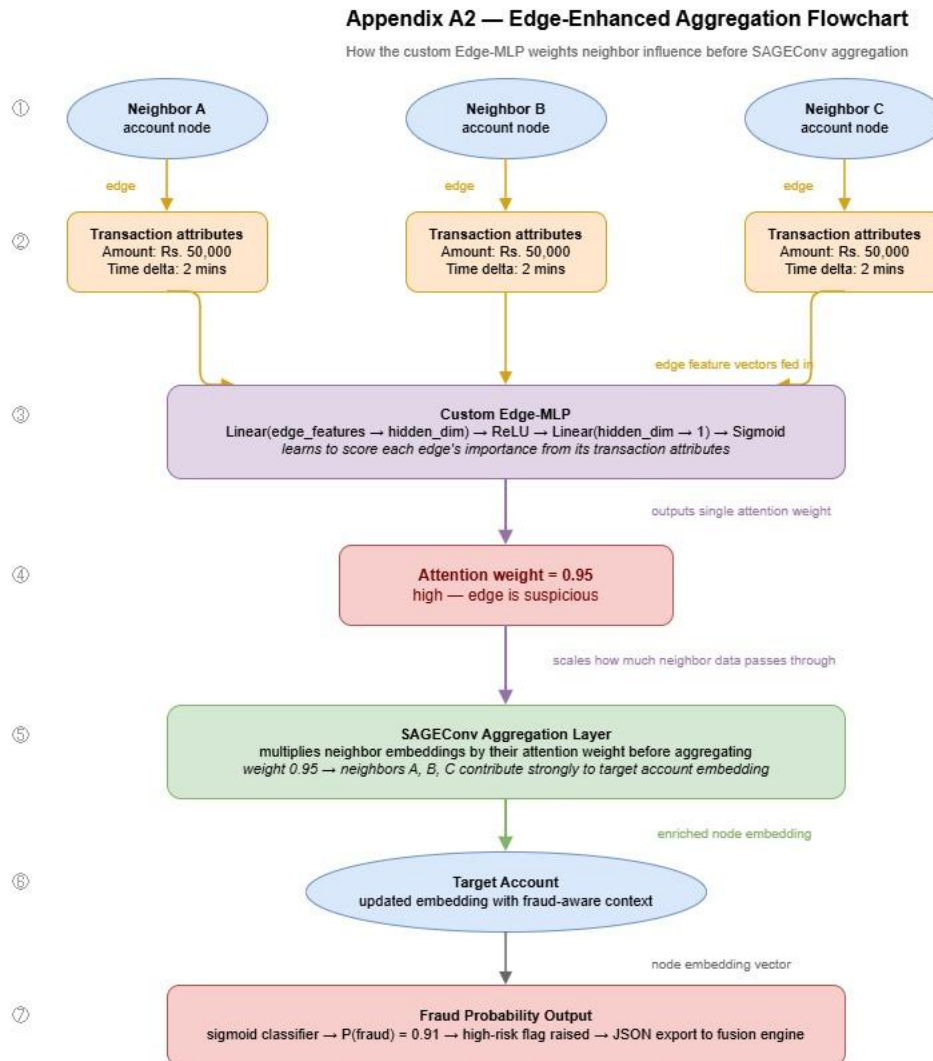
The following diagrams present the detailed technical architecture and data flow of the proposed system.



Appendix Figure A1. Detailed System Architecture Diagram



Appendix Figure 1.2 Detailed Data Flow Diagram

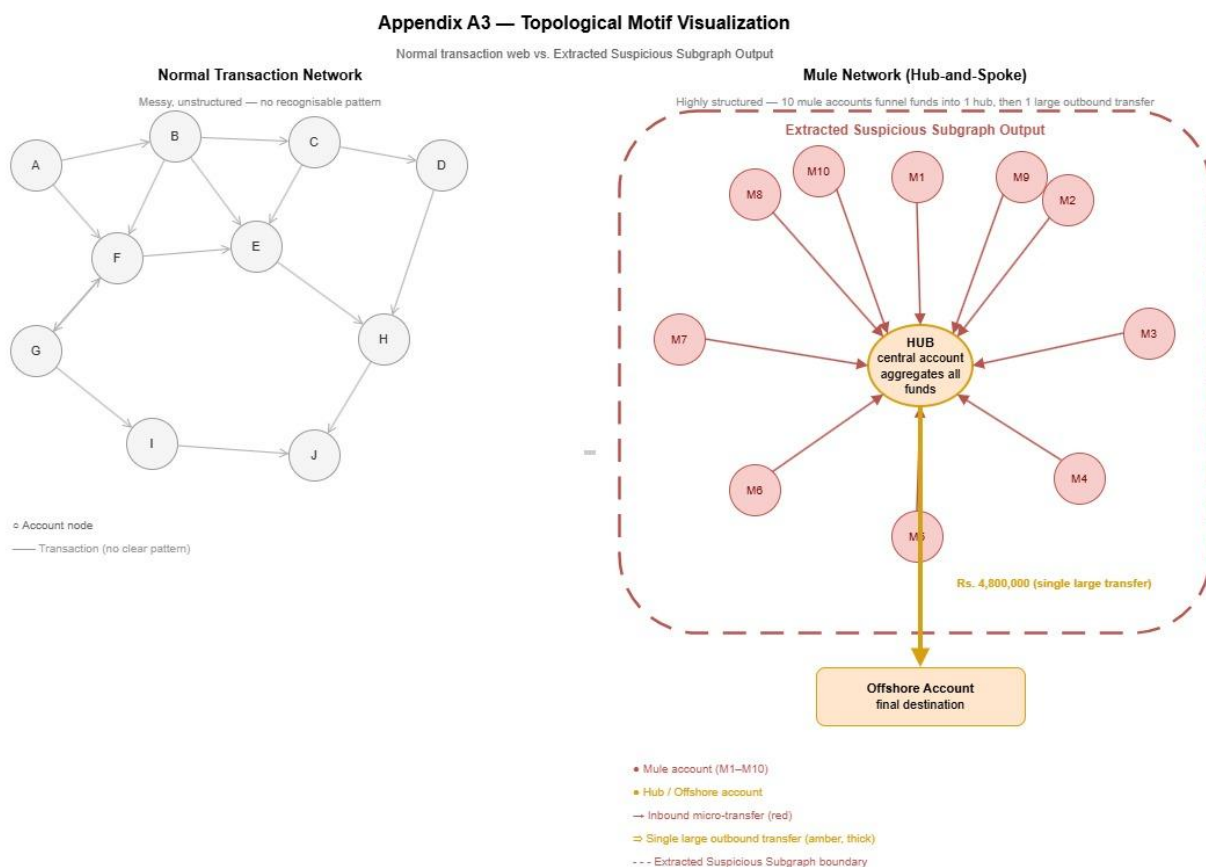


Appendix A2 – Edge Enhanced Aggregation Flowchart

10.3 Preliminary UI mockups

As the Edge Enhanced GraphSAGE Relational Fraud Detector functions strictly as a backend intelligence and mathematical engine, it does not possess a standalone graphical user interface. The visual output of this module is represented by the Topological Motif

Visualizations.



Appendix Figure A3. Topology motif Visualization

10.4 Risk analysis

Risk	Impact	Likelihood	Mitigation Strategy
Model predicting "0" for everything due to severe class imbalance	High	High	Abandon standard accuracy metrics. Implement and strictly monitor the Graph-Aware Imbalance Sampler

Out-Of-Memory (OOM) crashes during GraphSAGE training	High	High	Utilize dedicated Cloud GPU instances (Google Collab Pro).
High inference latency bottlenecking the main Fusion Engine	High	Medium	Develop the backend using asynchronous FastAPI.
Model overfitting to PaySim's synthetic artifacts	Medium	Medium	Apply feature dropout during training and ensure critical edge features
Integration failure due to mismatched JSON data payloads	High	Low	Establish a strict, documented API contract
Loss of pre-processed graph data or model weights	Medium	Low	Implement automated checkpoints during PyTorch training loops and sync all large tensor files to secure cloud backup storage after every successful epoch.

10.5 Ethical documents

- **Appendix Note 1 (Synthetic Data Usage Declaration)** - Confirmation that the GraphSAGE model is trained and evaluated exclusively utilizing the PaySim synthetic mobile money dataset. This ensures that no real-world Personally Identifiable Information (PII) or proprietary banking records are exposed during the research, completely mitigating GDPR and local financial secrecy risks.
- **Appendix Note 2 (Data Localization Security)** - Architectural guarantee that during enterprise deployment, all PyTorch Geometric tensor construction and graph processing occur locally behind the institution's firewall, preventing transaction metadata leakage.
- **Appendix Note 3 (Open-Source Compliance)** - Declaration that all underlying frameworks (PyTorch, PyTorch Geometric, NetworkX, FastAPI) are utilized under their respective open-source academic licenses.